

Cardiovascular Conditions

1. Pre-existing Cardiovascular Conditions

(These should be **patient inputs** — just like age, gender, mass, baseline measurements)

Condition	Why it matters as input	Likely coverage
Hypertension (with severity / controlled vs uncontrolled)	Changes baseline BP, vascular resistance, risk of end-organ damage	Engine-simulated (if Pulse supports) or Evidence-only
Coronary Artery Disease / Prior MI	Affects cardiac output, heart rate response, ischemia risk	Evidence-only or Engine-simulated
Heart Failure (HFrEF / HFpEF, NYHA class)	Directly impacts cardiac output, fluid handling, response to volume changes	Engine-simulated preferred
Prior CABG / PCI / Stents	Alters coronary flow reserve and response to stress	Evidence-only
Atrial Fibrillation / Arrhythmias	Affects heart rate, filling, stroke volume	Engine-simulated if supported
Valvular disease (AS, MR, etc.)	Changes pressure-volume relationships	Evidence-only / Unsupported for detailed modeling
Peripheral Arterial Disease	Affects systemic vascular resistance and tissue perfusion	Evidence-only
Congenital heart disease (if relevant)	Baseline anatomy/physiology different	Mostly Unsupported

2. Cardiovascular Diseases / Complications that Affect Simulation

(These can be triggered by interventions or evolve during the run)

- Acute Coronary Syndrome / Myocardial Ischemia
- Cardiogenic Shock
- Acute Heart Failure decompensation
- Hypertensive emergency / urgency
- Arrhythmias (VT, AF with RVR, bradycardia)
- Pulmonary Embolism (right-heart strain)
- Aortic dissection / aneurysm complications
- Cardiac tamponade
- Volume overload → worsening HF

3. Cross-Organ Effects (very important for your multi-agent design)

These are conditions where **cardiovascular state strongly influences other organs** (or vice-versa):

Cardiovascular state	Affects which organ agent	Mechanism
Low cardiac output / cardiogenic shock	Renal	↓ renal perfusion → AKI risk
High venous pressure / right-heart failure	Hepatic + Renal	Congestive hepatopathy, cardiorenal syndrome
Pulmonary hypertension / left-heart failure	Respiratory	Pulmonary edema, V/Q mismatch
Severe hypertension	Renal + Cerebral (if present)	Hypertensive nephropathy, stroke risk
Arrhythmias with low output	All organs	Global hypoperfusion
Fluid overload from HF	Respiratory + Renal	Pulmonary edema + worsening renal function

IN-SHORT

Cardiovascular inputs list (for simulation):

Pre-existing conditions we need as patient inputs:

- Hypertension (severity + controlled status)
- Heart Failure (type + NYHA)
- Prior CABG / PCI / MI
- CAD
- Atrial fibrillation / major arrhythmias
- Valvular disease (if any)

Cross-organ effects we must handle:

- Low cardiac output → Renal hypoperfusion / AKI
- Right-heart failure → Congestive hepatopathy + cardiorenal syndrome
- Left-heart failure → Pulmonary edema (Respiratory agent)

Coverage note: Many drug effects and exact ischemia trajectories will be **evidence-only**, not fully engine-simulated.